Merged and Enhanced Django + Python Web Development Syllabus

This document combines and enhances the contents of two syllabi: one focused on Python basics to Django transition, and the other focused on deep Django skills. It provides a full roadmap for both beginners and those advancing in web development.

1. Python Fundamentals

- • What is Python?
- • Variables, Data Types, and Operators
- • Control Structures (if, loops)
- • Functions and Modules
- • Lists, Tuples, Sets, Dictionaries
- • OOP in Python (Classes, Objects, Inheritance)
- • Exception Handling

2. Setting Up the Development Environment

- • Installing Python and Django
- • Setting up virtual environments using venv and pip
- • Choosing and configuring an IDE (VSCode, PyCharm, etc.)
- • Using .env for environment variables

3. Django Introduction and Project Setup

- • What is Django?
- • Django vs Other Web Frameworks
- • Understanding MVT Architecture
- • Creating Your First Django Project
- • Understanding Project and App Structure
- • Creating Django Apps
- • Django Admin Interface
- • Creating Superuser and Managing Models

4. Django Models and ORM

- • Defining Models and Fields
- • Field Types, Validators
- • Model Relationships (OneToOne, ForeignKey, ManyToMany)
- • Creating and Running Migrations
- • Using the Django Shell and QuerySet API
- • Using SQLite and Intro to PostgreSQL

5. Django Views and URL Routing

- • Function-based Views (FBVs)
- • Class-based Views (CBVs)

- • Generic Views (ListView, DetailView, etc.)
- • Mapping URLs using path() and re_path()

6. Templating with Django

- • Django Template Language (DTL)
- • Template Inheritance and Custom Tags
- • Creating Dynamic Web Pages
- • Working with Static Files (HTML, CSS, JS)
- • Using Bootstrap for Styling

7. Django Forms

- • Creating Forms using Form classes and ModelForms
- • Handling Form Submissions and Validations
- • Custom Validation and Error Handling
- • CSRF Protection
- • File Uploads with Forms

8. User Authentication & Authorization

- • Django's Built-in User Model
- • Login, Logout, Registration, Password Reset
- • Custom User Model (Optional, Advanced)
- • User Roles, Permissions, and Decorators
- • Protecting Views and Routes

9. Middleware, Sessions, and Signals

- • Understanding and Creating Middleware
- • Working with Cookies and Sessions
- • Using Django Signals for Decoupled Logic

10. Building RESTful APIs with DRF

- • Introduction to Django REST Framework
- • Serializers, Views, and ViewSets
- • Routers and URL Configuration
- • Handling Permissions and Authentication
- • Calling APIs from Frontend (Fetch, Axios)

11. Debugging, Testing, and Version Control

- • Django Debugging Tools
- • Writing Unit Tests (Models, Views, APIs)
- • Logging and Error Tracking
- • Using Git and GitHub for Version Control
- • Tools: ChatGPT, GitHub Copilot, Replit, Gemini, Blackbox

12. Deployment and Scaling

- • Preparing Django for Production
- • Static & Media Files (WhiteNoise, Cloudinary)
- • Deployment Platforms: PythonAnywhere, Railway, Render
- • Scaling Strategies: Horizontal and Vertical Scaling

13. Final Project and Showcase

- • Full-stack Django Web App (Blog, Job Portal, CRM, etc.)
- • CRUD Operations and Auth Integration
- • User Profiles and Dashboards
- • REST API for Selected Modules
- • Project Presentation and Peer Review
- • Contributing to Open Source and Career Networking